Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	12-10-2026
Team ID	SWUID2026402745
Project Name	HotelPro – Hotel Room Booking and Management System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements – HotelPro Room Booking & Management System

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Registration & Login (FR-REG)**	Allow a visitor to register by providing full name, email, phone number and password (FR-REG-01)
		Validate that email is unique and properly formatted and password meets minimum strength rules (8+ chars) (FR-REG-02)
		Hash passwords with a salted algorithm (bcrypt) before storing in MongoDB; prohibit plain-text storage (FR-REG-03)
		Allow registered users to log in with email and password and show clear error messages without revealing which field failed (FR-REG-04, 05)
		Allow authenticated users to log out, ending the session on the client (FR-REG-06)
		Allow users to view and update profile details and reset password via registered email (FR-REG-07, 08)
FR-2	**JWT Authentication & RBAC (FR-JWT)**	Generate a signed JWT containing user ID and role (customer, staff, admin) after successful login (FR-JWT-01)
		Configure JWT expiry time (e.g., 1 hour) signed with a secret key stored in an environment variable (FR-JWT-02)
		Send the JWT in the Authorization header using the Bearer scheme for all protected requests (FR-JWT-03)
		Verify token validity via authentication middleware and reject missing, expired or tampered tokens with HTTP 401 (FR-JWT-04)
		Enforce role-based access control (RBAC) via authorization middleware and return HTTP 403 when lacking permission (FR-JWT-05)
		Support refresh tokens to renew sessions without forcing re-login (FR-JWT-06)
FR-3	**Hotel & Room Management (FR-HRM)**	Allow an administrator to add a hotel with name, address, indexed city, description, amenities, images and rating (FR-HRM-01)
		Allow an administrator to edit, update or delete hotel details (FR-HRM-02)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Allow an administrator to add rooms to a hotel with room number, type (Single, Double, Deluxe, Suite), price per night, capacity, amenities and status (FR-HRM-03)
		Allow an administrator to update or remove a room and mark it as available, occupied, maintenance or inactive (FR-HRM-04)
		Prevent deletion of any room that has active or upcoming bookings (FR-HRM-05)
		Restrict all hotel and room management operations strictly to the Administrator role (FR-HRM-06)
FR-4	**Room Search & Availability (FR-SRC)**	Allow users to search rooms by city/hotel, check-in date, check-out date and number of guests (FR-SRC-01)
		Allow filtering and sorting of search results by price per night, room type, amenities and hotel rating (FR-SRC-02)
		Display only rooms that are not booked for any part of the selected date range (FR-SRC-03)
		Display full room details including photos, price per night, maximum capacity and amenities (FR-SRC-04)
		Reject searches where check-out date is not after check-in date or check-in date is in the past (FR-SRC-05)
		Return room search and availability results in under 2 seconds under normal load (FR-SRC-06)
FR-5	**Hotel Room Booking (FR-BKG)**	Allow a logged-in customer to book an available room for selected check-in and check-out dates (FR-BKG-01)
		Calculate total amount automatically as price per night multiplied by number of nights, including applicable taxes (FR-BKG-02)
		Re-verify room availability at the exact time of confirmation to prevent double booking (FR-BKG-03)
		Record primary and accompanying guest details (name, contact, number of adults/children) with the booking (FR-BKG-04)
		Generate a unique booking ID and set initial booking status to Pending until payment is completed (FR-BKG-05)
		Display a booking summary and confirmation on screen upon success and send a confirmation email (FR-BKG-06, 07)
FR-6	**Booking Management (FR-BMG)**	Allow customers to view their current, upcoming and past reservations in My Bookings (FR-BMG-01)
		Allow customers to cancel a booking before check-in, subject to the cancellation policy (FR-BMG-02)
		Allow customers to modify booking dates where the room is available for the new dates (FR-BMG-03)
		Allow hotel staff and administrators to view, search and filter all bookings by status, date or guest (FR-BMG-04)
		Maintain booking statuses: Pending, Confirmed, Checked-in, Checked-out and Cancelled (FR-BMG-05)

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		Automatically make cancelled rooms available again for the freed dates (FR-BMG-06)
FR-7	**Guest Management (FR-GST)**	Allow hotel staff to add, view, edit and search guest records (name, contact, address, ID proof type and number) (FR-GST-01)
		Link each guest record to one or more bookings in the database (FR-GST-02)
		Allow staff to record additional accompanying guests alongside the primary guest (FR-GST-03)
		Protect guest personal data and restrict access exclusively to authorized staff and administrators (FR-GST-04)
		Maintain a complete guest stay history across reservations (FR-GST-05)
FR-8	**Check-in & Check-out (FR-CIO)**	Allow staff to check in a guest against a confirmed booking and update booking status to Checked-in (FR-CIO-01)
		Automatically set the room status to Occupied upon guest check-in (FR-CIO-02)
		Allow staff to check out a guest, update booking to Checked-out and set room to Available or Cleaning/Maintenance (FR-CIO-03)
		Verify that payment is completed or record the outstanding balance at check-out (FR-CIO-04)
		Record actual check-in (actualCheckIn) and actual check-out (actualCheckOut) date and time (FR-CIO-05)
		Calculate extra charges for early check-in or late check-out when applicable (FR-CIO-06)
FR-9	**Payment Processing (FR-PAY)**	Allow customers to pay online using supported methods (credit/debit card, UPI, net banking) via an integrated payment gateway (FR-PAY-01)
		Prohibit storing card numbers, CVV or sensitive card data in the system (FR-PAY-02)
		Record payment amount, currency (INR), method, unique transaction ID, status (Pending, Success, Failed, Refunded) and timestamp (FR-PAY-03)
		Update booking status to Confirmed strictly after a successful payment (FR-PAY-04)
		Support full or partial refunds for eligible cancellations and record refund transactions (FR-PAY-05)
		Generate a payment receipt/invoice for each successful payment and allow staff to record offline cash/counter payments (FR-PAY-06, 07)
FR-10	**Admin Dashboard & Reports (FR-ADM)**	Display key metrics: total bookings, active bookings, cancellations, occupancy rate, available rooms and total revenue (FR-ADM-01)
		Provide visual charts and summaries aggregated by day, week and month (FR-ADM-02)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Allow administrators to manage users, including changing roles (customer, staff, admin) and activating/deactivating accounts (FR-ADM-03)
		Allow administrators to view, search and filter all bookings and payment records (FR-ADM-04)
		Allow reports to be exported in CSV or PDF format (FR-ADM-05)
		Restrict dashboard access strictly to users with the Administrator role (FR-ADM-06)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The UI shall be intuitive so a first-time user can complete a booking in a few steps without training, built as a responsive React.js and Bootstrap SPA supporting desktop, tablet and mobile screens, with inline form validation, clear feedback messages and basic accessibility (readable contrast, labels, keyboard navigation).
NFR-2	**Security**	All communication over HTTPS/TLS; passwords hashed using bcrypt with a suitable salt factor; JWT secrets and database credentials stored in environment variables; server-side RBAC on all protected endpoints; input validation and sanitisation against NoSQL injection, XSS and CSRF; rate limiting on authentication endpoints; zero sensitive card storage; security event logging.
NFR-3	**Reliability**	Booking and payment operations shall be consistent so that a failed payment never results in a confirmed booking and no room is double-booked; graceful error handling returning meaningful error messages and standard HTTP status codes (200, 201, 400, 401, 403, 404, 409, 500).
NFR-4	**Performance**	Web pages shall load within 3 seconds on a standard broadband connection; API responses for common operations (login, search, booking) shall return within 2 seconds for 95% of requests under normal load; supports at least 500 concurrent users without degradation; MongoDB queries optimised with indexes on email, hotelId, roomId, city, transactionId and compound (roomId, checkInDate, checkOutDate).
NFR-5	**Availability**	The system shall target 99% availability (excluding planned maintenance), supported by regular automated backups of the MongoDB database with the ability to restore within a defined recovery time.
NFR-6	**Scalability**	Stateless JWT-based Node.js/Express.js REST API allows horizontal scaling of back-end servers behind a load balancer; MongoDB deployable as a replica set or sharded cluster on MongoDB Atlas to handle data and traffic growth; modular architecture allows new hotels, branches and features to be added without major redesign.

NFR No.	Non-Functional Requirement	Description
NFR-7	**Maintainability**	Modular codebase structure (routes, controllers, Mongoose models, middleware, services) following consistent naming, formatting and commenting conventions; version control tracked with Git and GitHub branches/pull requests; RESTful APIs documented and tested using Postman collections.
NFR-8	**Compatibility & Portability**	Works across the latest two versions of Google Chrome, Mozilla Firefox, Microsoft Edge and Apple Safari on desktop, tablet and smartphone; back end runs on Windows 10/11, Linux (Ubuntu 20.04+) and macOS with Node.js LTS; uses standard JSON over REST so future mobile or third-party clients can integrate.